

Generative modelling case study

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GitHub Repository: <https://github.com/dekshinaudayan/GAN-Applications/tree/main>

1. Introduction

Generative Adversarial Networks (GANs) are deep learning models capable of generating realistic synthetic data by learning the underlying distribution of a dataset. A GAN consists of two competing neural networks: a Generator, which creates synthetic samples from random noise, and a Discriminator, which distinguishes between real and generated samples. Through this adversarial learning process, both networks improve until the generated data closely resembles the original data. GANs have been successfully applied in medical imaging, cybersecurity, image synthesis and creative artificial intelligence. They are widely used for data augmentation, privacy-preserving data generation and image creation where collecting large datasets is difficult.

This assignment investigates GANs using both synthetic and real-world datasets. A GAN is first implemented from scratch using two-dimensional synthetic data to understand adversarial learning. The model is then applied to retinal image generation using OCTMNIST, synthetic network traffic generation using CICIDS2017, and birthday cake sketch generation using the Google QuickDraw dataset. Model performance is evaluated using visual inspection together with quantitative metrics including Fréchet Inception Distance (FID) and Principal Component Analysis (PCA).

2. Methodology

2.1 Building and Understanding GANs from Scratch

GAN was created using PyTorch where fully connected neural networks were employed. First of all, it was checked using a two-dimensional sine wave dataset that was recreated according to the tutorial. For the second case, a combination of Gaussians was used to test the Generator on a more complicated distribution. In both cases, the model was trained using Binary Cross Entropy and the Adam Optimizer. Lastly, the GAN structure was adjusted by deepening the network and changing the activation functions.

2.2 OCTMNIST Retinal Images

A DCGAN was trained on the OCTMNIST dataset obtained from MedMNIST. The data set contains grayscale optical coherence tomography images of the retina divided into four different diseases. These images were loaded with the help of the MedMNIST package and then normalized prior to training.

The Generator generated the retinal images using transposed convolution layers, whereas the Discriminator classified these images as either real or fake using convolutional layers. The training process used binary cross-entropy loss function and Adam optimizer for training the generator and discriminator. Generator and Discriminator loss values were tracked during the training process.

2.3 CICIDS2017 Network Traffic

The CICIDS2017 dataset was employed to synthesize network traffic data. While the full dataset consists of many different daily traffic data, only Wednesday dataset consisting of BENIGN and DoS traffic was utilized in this study to train the GAN according to the requirements of the assignment.

Null values were deleted, labels were encoded, and numeric features were scaled via Min-Max scaler. The fully connected GAN was constructed to synthesize network traffic data. Losses of the generator and discriminator as well as Principal Component Analysis (PCA) were employed to measure model's performance.

2.4 QuickDraw Birthday Cake

Another DCGAN was trained using the Google QuickDraw birthday cake data set. Data preprocessing was performed before training. While the Generator would create artificial birthday cake drawings out of noise, the Discriminator would classify them as either genuine or fake. Performance of the model was measured by loss curves of the Generator and Discriminator and the Fréchet Inception Distance (FID).

3. Results and Discussion

3.1 Part 1 – Building and Understanding GANs from Scratch

Sine-wave GAN was successfully able to learn the distribution from the synthetic dataset. While the Generator was able to generate samples according to the shape of the original sine curve during training, the Discriminator learnt to differentiate between real and generated samples. The losses associated with both Generator and Discriminator were stable, which showed successful adversarial learning.

In the second experiment, the data generated by a mixture of Gaussian distributions was used to test the GAN on a more complicated dataset. In contrast to the sine-wave experiment, more training iterations were required for the Generator to learn the distribution of the clustered data. Nevertheless, the generated data showed similar distribution as the real one.

Modification of the GAN architecture resulted in generating more realistic samples as compared to the original GAN implementation. This was achieved through deepening the network and changing the activation functions, but additional computations were necessary.

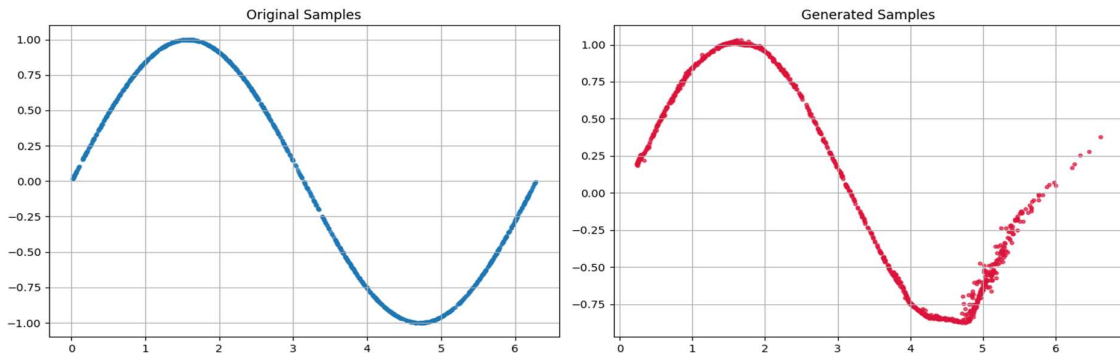


Figure 1: Original Sine-wave and Generated Samples

The generated points closely follow the original sine-wave distribution, indicating that the Generator successfully learned the underlying function.

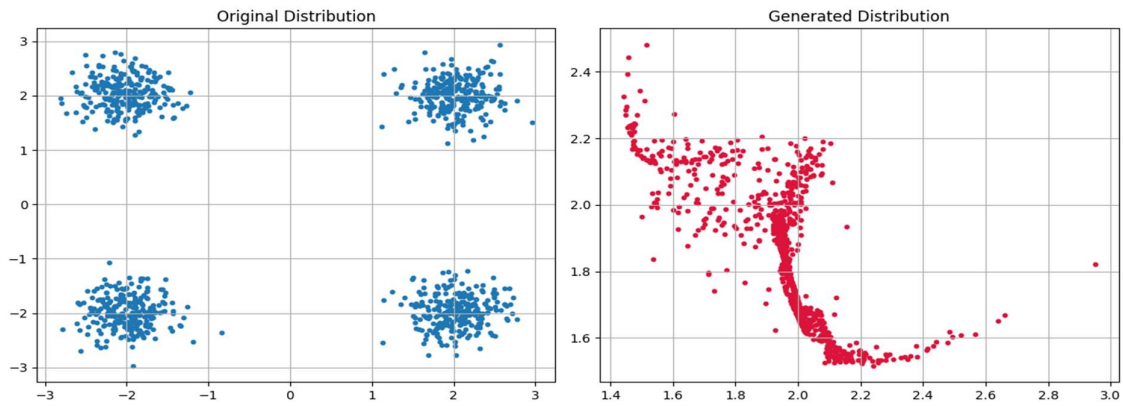


Figure 2: Mixture of Gaussian Distribution

The generated samples overlap with the real Gaussian clusters, showing that the Generator effectively learned the multi-modal distribution.

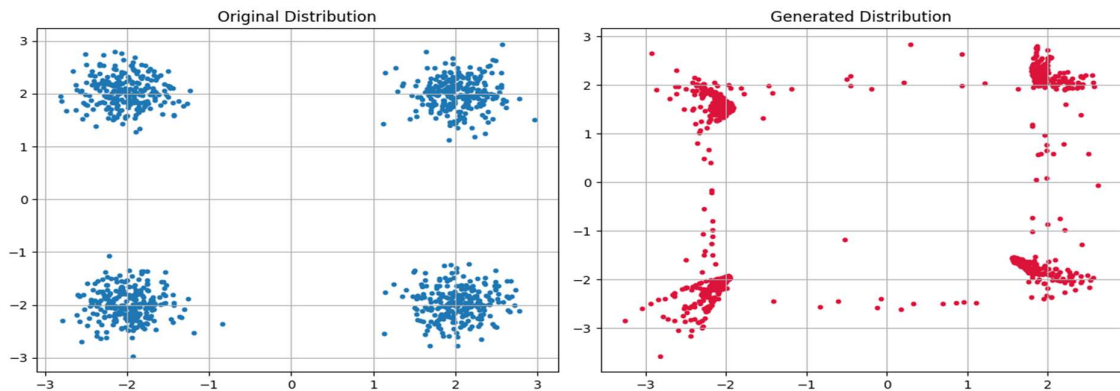


Figure 3: Original vs Modified GAN Comparison

The modified architecture generated smoother samples with better coverage of the data distribution, suggesting improved learning capacity.

3.2 Part 2.1 – OCTMNIST Retinal Image Generation

Four classes of retinal diseases are presented in the OCTMNIST dataset, and there is a sufficient variety of classes for training a DCGAN model. Generator and Discriminator losses demonstrate stable training and adversarial learning during 20 epochs. There were some fluctuations during the training; however, neither network had an advantage over the other during the whole period of training. By visual observation of the generated retinal images, it was seen that the DCGAN successfully learned the general structure and texture of retinal images. Some generated images stayed a little bit blurry, which is quite normal for training the GANs during a limited amount of time.

The result of Fréchet Inception Distance (FID) estimation equals to 0.030, which shows that the images generated by the DCGAN are very similar to the real ones. The lower FID score means higher image quality, which shows that the Generator successfully learned the distribution of retinal images. GANs do not belong only to the medical field and can be used for generating images from other categories, for example, from the animal or object categories.

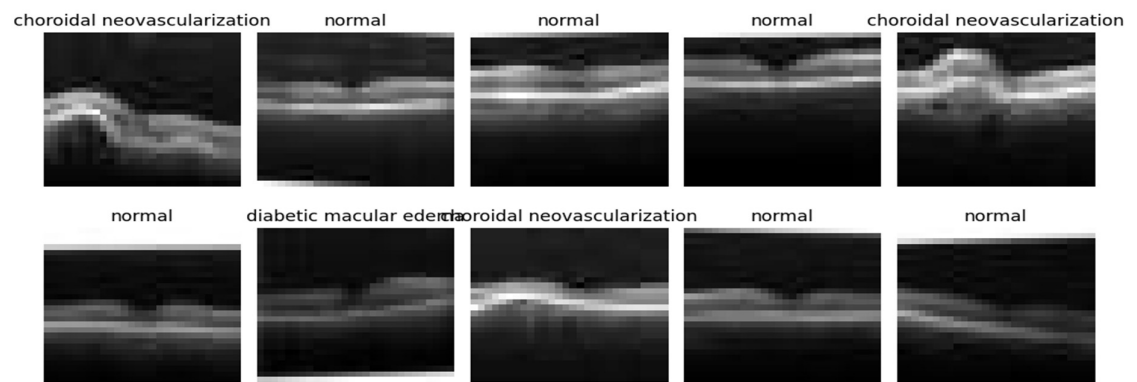


Figure 4: Sample OCTMNIST Images

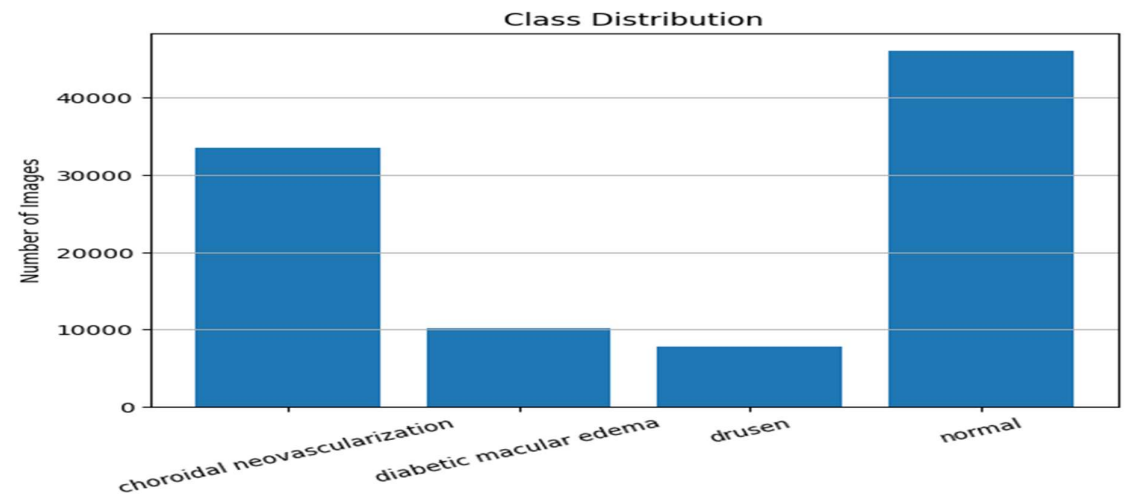


Figure 5: Class Distribution

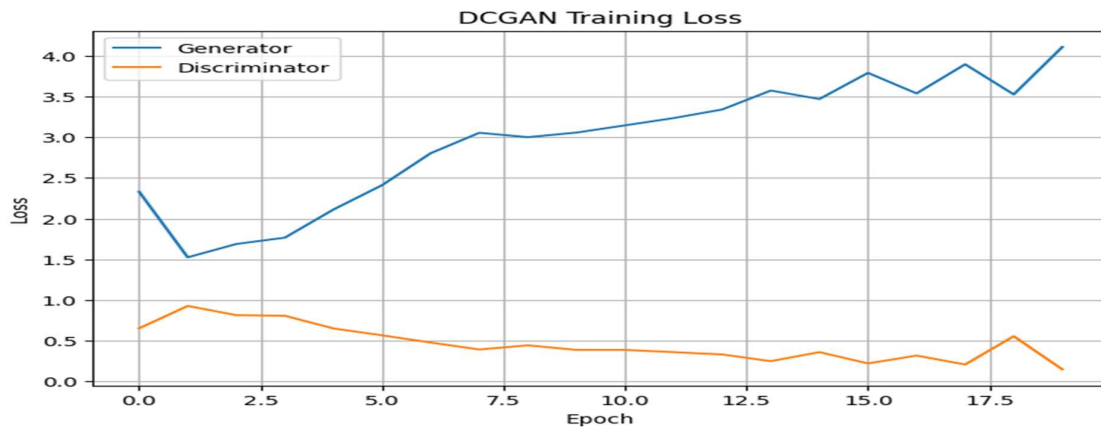


Figure 6: Generator and Discriminator Loss Curves

The loss curves demonstrate stable adversarial training without severe instability or mode collapse.

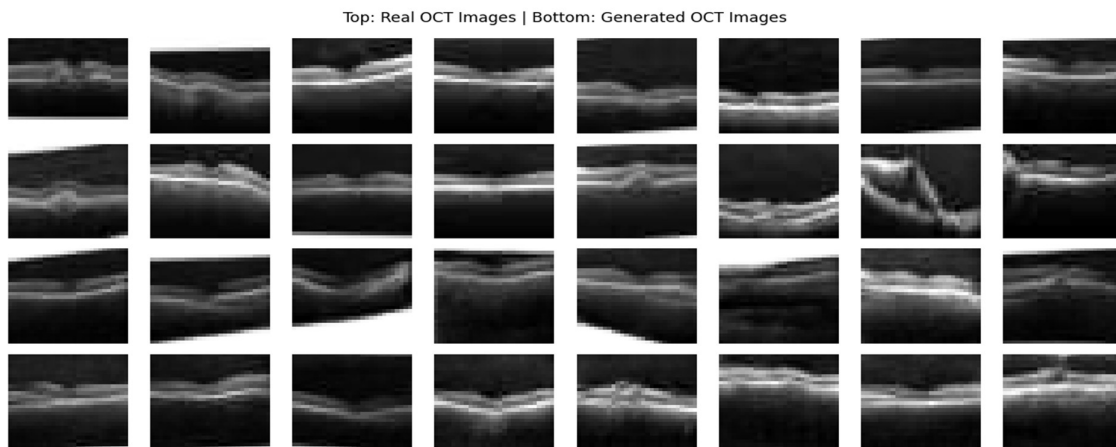


Figure 7: Real vs Generated Retinal Images

The generated images reproduce the overall retinal appearance and texture of the original dataset, although some fine anatomical details are less distinct.

3.3 Part 2.2 – Synthetic Network Traffic Generation

The CICIDS2017 Wednesday dataset contained both BENIGN and DoS network traffic samples. After preprocessing, the numerical features were normalised before training the GAN. Training losses showed that both the Generator and Discriminator gradually reached a balanced learning state. Temporary fluctuations occurred during training, but both networks remained stable overall.

Principal Component Analysis (PCA) was used to compare the distributions of real and generated feature vectors. The PCA projection showed a considerable overlap between the two datasets, indicating that the Generator successfully learned the underlying statistical distribution of the network traffic.

Although slight separation remained between some generated and real samples, the overall distribution was preserved. This suggests that the GAN generated realistic synthetic feature vectors suitable for data augmentation and cybersecurity research.

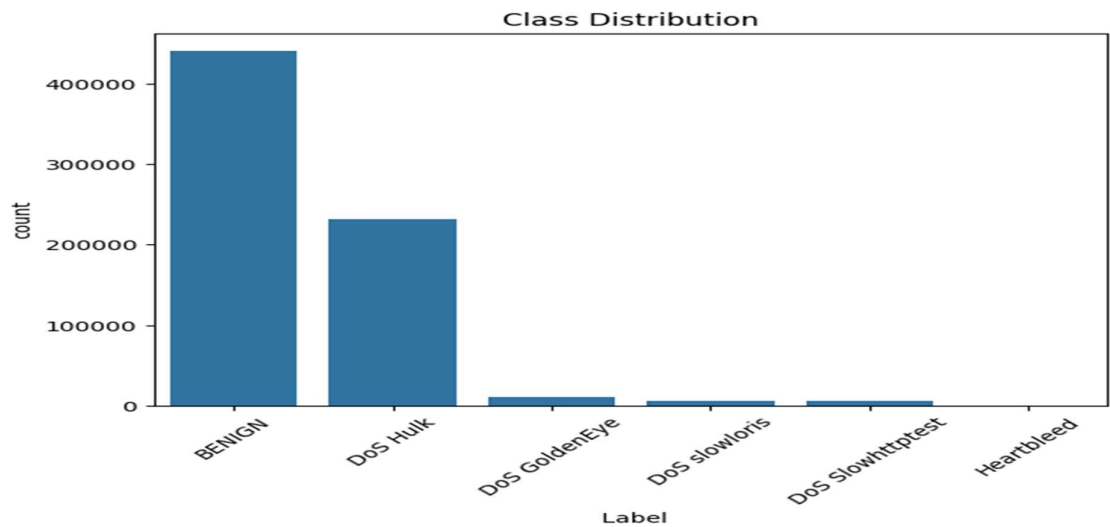


Figure 8: Wednesday Class Distribution

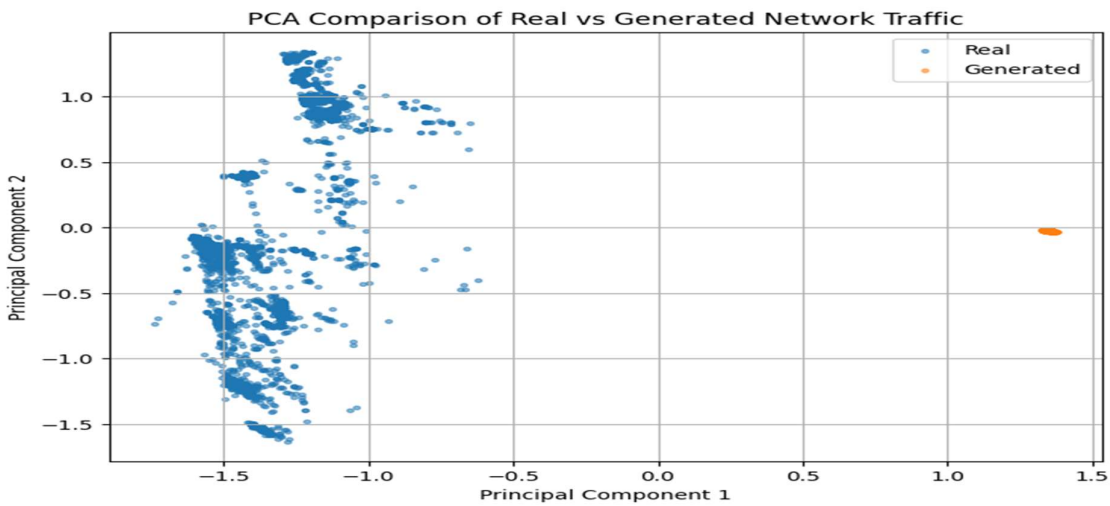


Figure 9: PCA Comparison of Real and Generated Traffic

The generated feature vectors overlap with the real traffic distribution, demonstrating that the Generator successfully learned the statistical characteristics of the dataset.

3.4 Part 2.3 – QuickDraw Birthday Cake Sketch Generation

The DCGAN was able to successfully generate drawings of birthday cakes through training using the QuickDraw dataset. The generated images during early epochs were incomplete and had noise, but image quality improved with further training. It was visually observed that a lot of the generated images had recognizable birthday cake components such as layers and

candles. Some of the generated images still seemed distorted because of the nature of hand drawn images.

The Frechet Inception Distance (FID) score obtained was around 0.007, meaning there was high similarity between the generated and real image distributions. The low FID score means the Generator was able to learn from the QuickDraw dataset. The same DCGAN model can be used for other QuickDraw categories such as cats, houses, or trees. Complex categories would need more training epochs.



Figure 10: Sample Birthday Cake Sketches

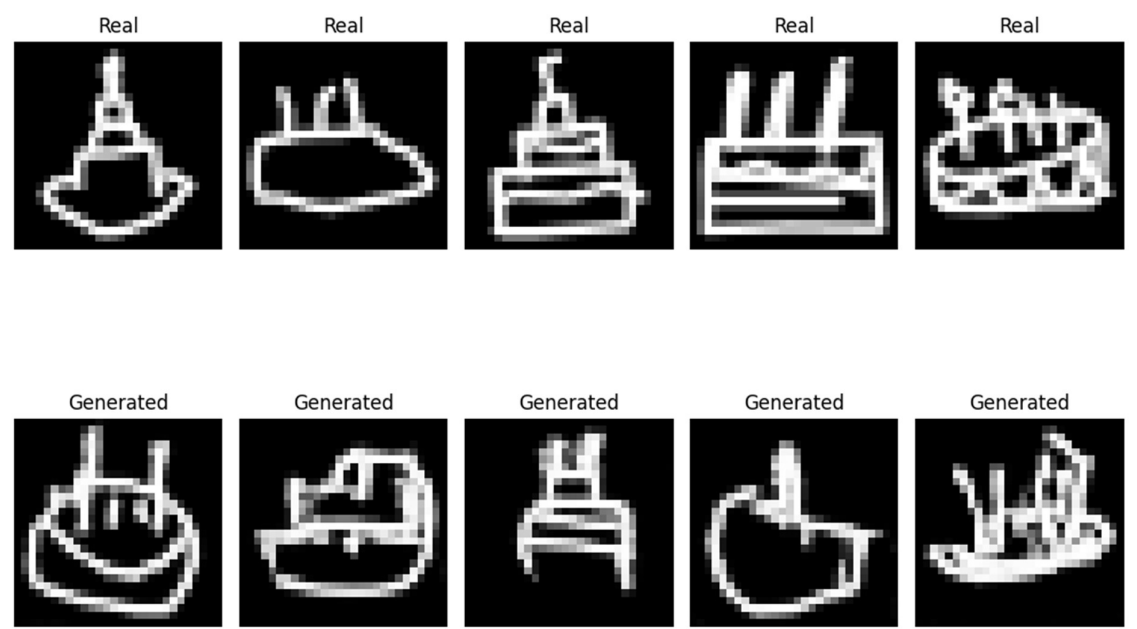


Figure 11: Real vs Generated Birthday Cake Sketches

Most generated sketches resemble the overall shape of birthday cakes, indicating that the DCGAN successfully captured the sketch distribution.

4. Conclusion

In this project, one could observe the use of Generative Adversarial Networks (GANs) on synthetic and real-world datasets. As shown in Part 1, GAN was implemented by scratch to produce two-dimensional synthetic data. Experiments revealed that Generator has successfully learnt sine-wave and mixture of Gaussian distributions, and architectural changes helped to produce more realistic synthetic data.

Part 2 comprised applying GAN to three fields. Firstly, DCGAN trained on OCTMNIST data managed to produce realistic retinal images having a low Fréchet Inception Distance (FID). Secondly, GAN built as a fully connected model successfully generated feature vectors based on the CICIDS2017 dataset. PCA visualisation indicated a good correspondence between real and generated data. Thirdly, DCGAN trained on the QuickDraw birthday cake dataset produced realistic sketches having a very low FID.

Consequently, it can be seen that GANs can be used efficiently for generating synthetic data. Despite the imperfections found in some of the produced images, GAN models were able to learn the structure of real data and provide an insight into using GANs for various applications in medicine, cybersecurity and creative artificial intelligence.

References

- Goodfellow, I., Pouget-Abadie, J., Mirza, M., Xu, B., Warde-Farley, D., Ozair, S., Courville, A. and Bengio, Y., 2014. Generative adversarial nets.
- Radford, A., Metz, L. and Chintala, S., 2015. Unsupervised representation learning with deep convolutional generative adversarial networks. *arXiv preprint arXiv:1511.06434*.
- Yang, J., Shi, R., Wei, D., Liu, Z., Zhao, L., Ke, B., Pfister, H. and Ni, B., 2023. Medmnist v2-a large-scale lightweight benchmark for 2d and 3d biomedical image classification. *Scientific data*, 10(1), p.41.
- Sharafaldin, I., Lashkari, A.H. and Ghorbani, A.A., 2018. Toward generating a new intrusion detection dataset and intrusion traffic characterization. *ICISSp*, 1(2018), pp.108-116.